



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Effect sizes

Understanding, choosing and computing the
currency of meta-analysis

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Learning objectives

By the end of this session, you will be able to:

- 1 Explain what an **effect size** is and why each one comes with a **sampling variance**.
- 2 **Choose** the right metric for your data: mean difference, Hedges' g , log response ratio, log odds ratio, Fisher's z .
- 3 **Compute** them by hand and with `metafor::escalc()`.
- 4 Recognise the **traps**: small samples, near-zero means, missing SDs, unsafe conversions.

What is an effect size?

Studies do not speak the same language

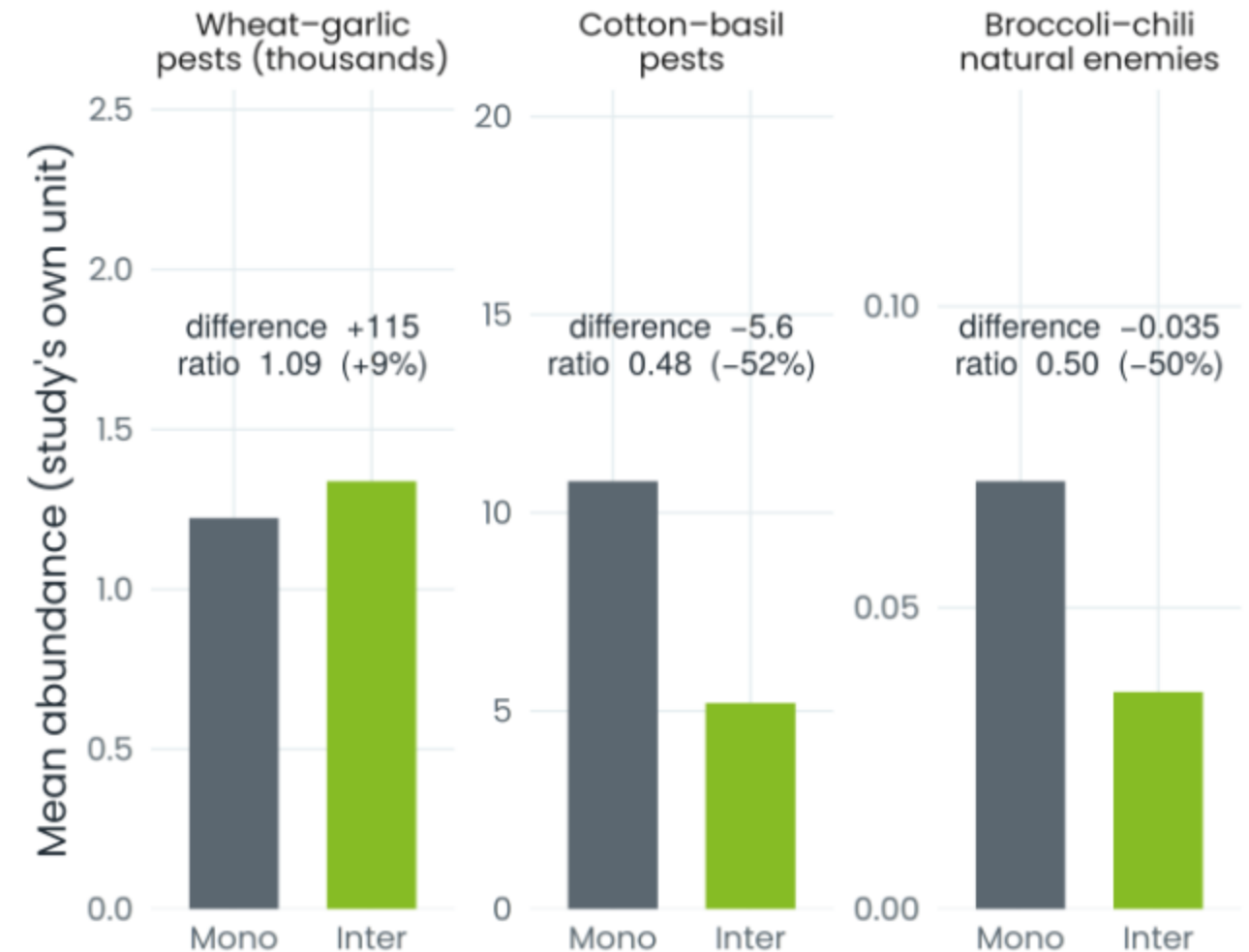
Each of the 888 comparisons gives a **mean in the intervention** (intercropping) and a **mean in the comparator** (monoculture), each in the study's own unit: per trap, per plant, per m²...

The raw differences (+115, -5.6, -0.035) cannot be compared. The **ratio** can: two studies halve the abundance, one barely changes it.

An effect size

compares the **intervention mean with the comparator mean on one common scale** (direction and magnitude), and comes with its **uncertainty**.

Three comparisons, three scales



Source: three comparisons from Jones et al. (2021).

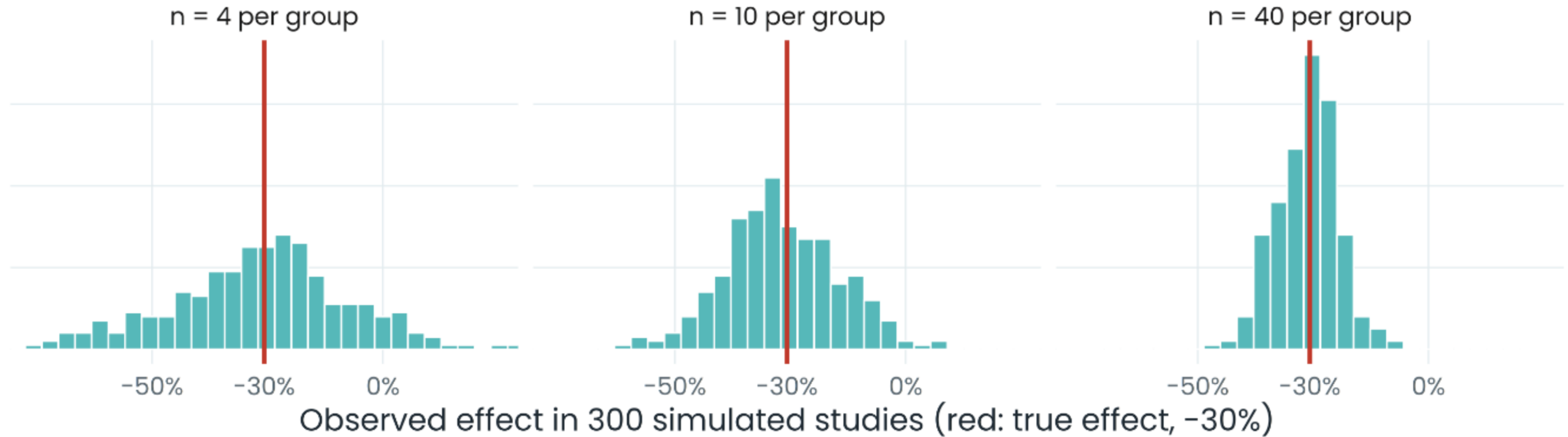
Why not just use p-values?



A p-value mixes **effect size** and **sample size**: a tiny effect can be “highly significant”, a large one “not significant”.

Meta-analysis combines **effect sizes and their uncertainty**, not p-values (Nakagawa and Cuthill 2007).

Every study sees the truth through noise



$$\hat{\theta}_k = \theta_k + \varepsilon_k \quad \text{observed} = \text{true} + \text{sampling error}, \quad \text{Var}(\varepsilon_k) = v_k$$

Each effect size comes with its **sampling variance** v_k . In the meta-analysis this afternoon, each study is **weighted** by the inverse of its variance, $w_k = 1/v_k$: precise studies count more. (In a random-effects model, the weight also includes the real variation between studies, τ^2 : $w_k = 1/(v_k + \tau^2)$.)

Which metric for which data?

A decision tree

Continuous outcome, two groups

means, SD, n

- Same unit everywhere → **Mean difference (MD)**
- Scores, indices, values that can be negative → **standardised mean difference** (Hedges' g)
- Quantities: counts, biomass, yield → **log response ratio (lnRR)**

Binary outcome, two groups

counts in a 2×2 table

- Presence/absence, survival, infestation → **log odds ratio** (or log risk ratio)

Relationship between two continuous variables

correlation r, n

- e.g. yield vs landscape complexity → **Fisher's z**

Test: does “**twice as many**” make sense?

Yes (aphids, yield) → **lnRR**.

No → **g**: damage scores, soil quality indices, assays that differ between studies, values that can be negative (change in soil C stock).

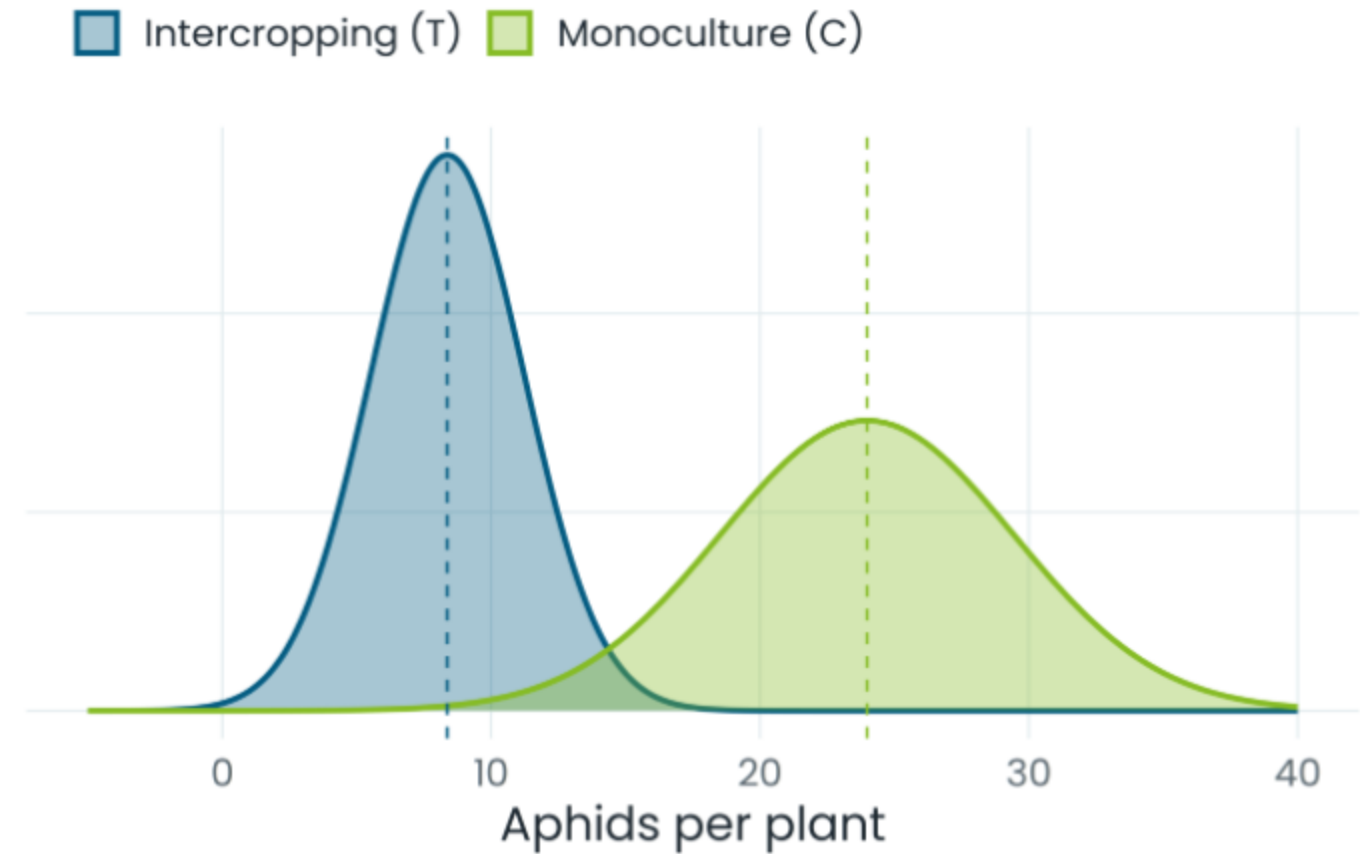
Our running example → **lnRR**. Fix the metric **in the protocol**, before seeing the results.

Show of hands: which metric for your own data?

Our worked example

Aphids on fennel, **intercropped with cotton** vs **fennel monoculture** (Fernandes et al. 2013), one comparison of our dataset:

	Intercropping (T)	Monoculture (C)
Mean	8.37	23.98
SD	2.85	5.46
n	4	4



Convention for the whole week: **T (intervention) in the numerator / first**, C (comparator) second. A negative effect = fewer pests with intercropping.

The log response ratio (lnRR)

$$\ln RR = \ln \left(\frac{\bar{X}_T}{\bar{X}_C} \right)$$
$$v_{\ln RR} = \frac{SD_T^2}{n_T \bar{X}_T^2} + \frac{SD_C^2}{n_C \bar{X}_C^2}$$

(Hedges et al. 1999)

Back to percent for reporting:

$$\text{change} = 100 \times (e^{\ln RR} - 1) \%$$

Why the log?

The ratio is **asymmetric** (halving = 0.5, doubling = 2); its log is **symmetric** (-0.69, +0.69) and closer to normal.

Why ecologists like it

Unit-free and **proportional**: “intercropping reduces aphids by 65%” speaks to agronomists — and its value does not depend on the SD (only its variance does).

Exercise 1 · By hand (pairs, 8 min)

CALCULATOR READY

With the aphid example (T: 8.37 ± 2.85 , $n = 4$; C: 23.98 ± 5.46 , $n = 4$), compute lnRR, its variance, and the % change.

$$\ln RR = \ln(8.37 / 23.98) = -1.053$$

$$v = 2.85^2 / (4 \times 8.37^2) + 5.46^2 / (4 \times 23.98^2) = 0.0419 \text{ (SE} = 0.205\text{)}$$

$$\text{Change} = 100 \times (e^{-1.053} - 1) = -65\%$$

Now let R do it (live coding, 5 min)

PREDICT FIRST What value of y_i should R return for our aphid example?

```
1 library(metafor)
2 # ic <- read.csv("intercropping_biodiversity.csv")      # running example
3
4 dat <- escalc(measure = "ROM",                        # lnRR ("ratio of means")
5               m1i = b_mean_t, sd1i = b_sd_t, n1i = b_n_t, # T = intercropping
6               m2i = b_mean_c, sd2i = b_sd_c, n2i = b_n_c, # C = monoculture
7               data = ic)
8 # other metrics: measure = "SMD" (Hedges' g), "OR", "ZCOR"
9
10 dat[dat$es_id == 2011, c("es_id", "yi", "vi")]        # our worked example
```

```
   es_id    yi    vi
137 2011 -1.0526 0.0419
```

```
1 c(NA_yi = sum(is.na(dat$yi)), vi_zero = sum(dat$vi == 0, na.rm = TRUE)) # what to deal with
```

```
NA_yi vi_zero
156      46
```

- y_i = effect size, vi = sampling variance: **the input of every meta-analysis** (`rma()`, this afternoon).
- Zero mean → NA; zero SD → $vi = 0$ ("infinite" precision!): **count both** before `rma()`.

Hedges' g , step by step

Idea: express the difference in **standard-deviation units**.

$$d = \frac{\bar{X}_T - \bar{X}_C}{SD_{pooled}}$$

Useful when studies use **different scales** (scores, indices), where “twice as many” means nothing.

Small samples **inflate** d → multiply by a correction $J < 1$: that is **Hedges' g** (Hedges 1981).

With our aphids:

1. Difference: $8.37 - 23.98 = -15.61$ aphids
2. Pooled SD (equal n : average the variances):
 $\sqrt{((2.85^2 + 5.46^2) / 2)} = 4.36$
3. $d = -15.61 / 4.36 = -3.58$ standard deviations
4. With $4 + 4$ plants, $J = 0.87 \rightarrow g = -3.12$

`escalc(measure = "SMD")` does all of this — and the variance. You never compute g by hand in practice.

The traps

Trap 1 • Means close to zero

YOUR CALL

156 comparisons in our data have a zero mean.
What do you do with them for a InRR?

Exact zeros: no InRR (and most also have $SD = 0$, so no g). Report them; test the conclusion **with and without**.

Near zero: InRR is biased. Geary's rule: each mean $\geq 3 \text{ SE}$ from zero $\rightarrow$ **28%** of our usable comparisons fail it.

What to do

- Near zero or small n : **bias-corrected InRR** (Lajeunesse 2015) — formula and R code on the handout.
- **Never** add an arbitrary constant (+1) without testing its influence.

Trap 2 · Missing or disguised SDs

The common case: a standard error

$$SD = SE \times \sqrt{n}$$

553 of our 888 comparisons were reported with a SE.

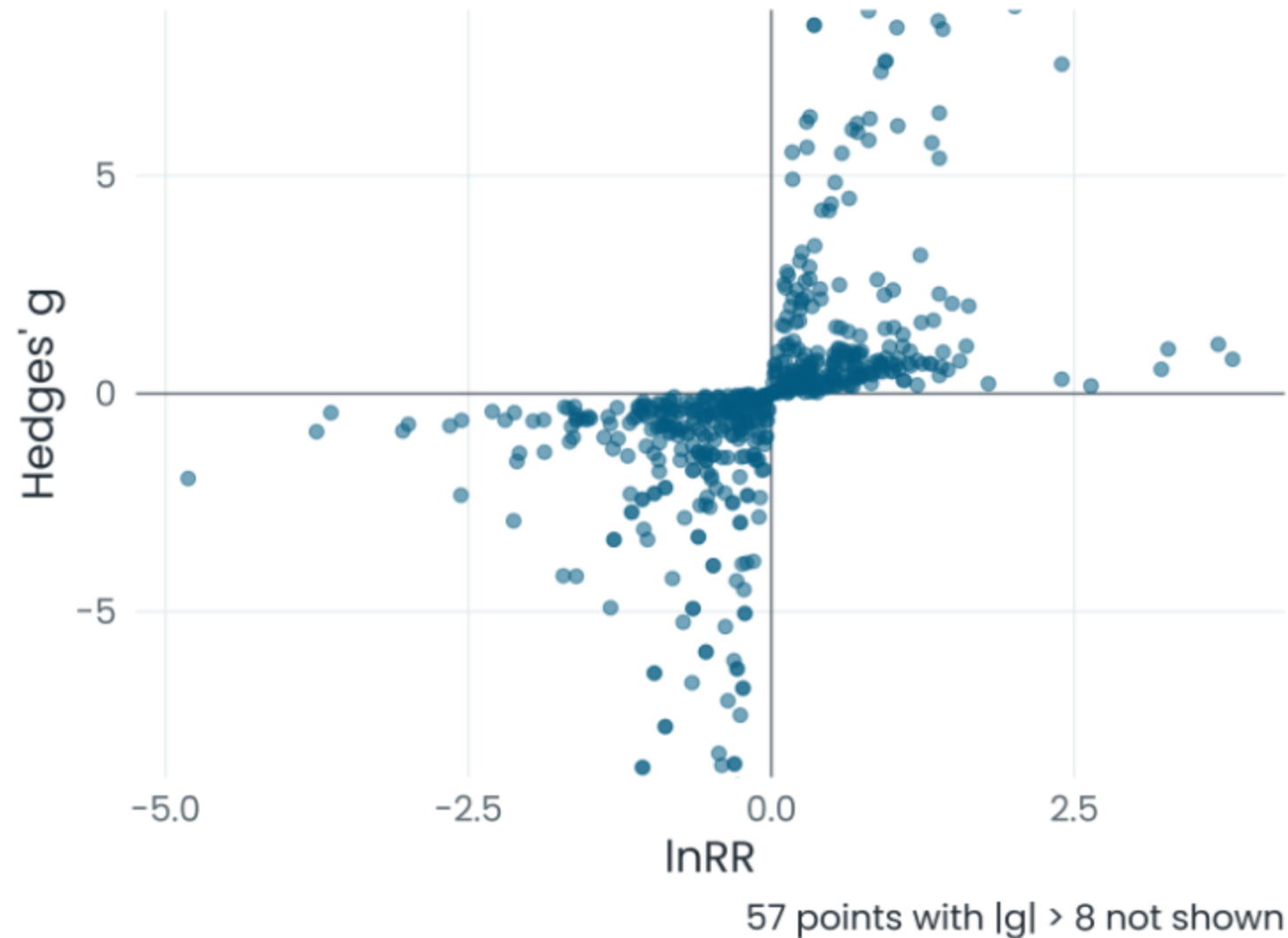
Also possible: from a 95% CI, or from a median and IQR (Wan et al. 2014) — see the formula sheet.

215 comparisons (**24%**) have **no usable SD**.

Never set it to 0 or drop silently. Robust fix for lnRR: compute the sampling variance from the **typical coefficient of variation** of the other studies ($CV = SD/\text{mean}$, averaged with weights n , then squared) (Nakagawa et al. 2023), after screening implausible CVs (Wednesday 16:15). Not possible for zero means.

Traps 1 and 2 together: **224** of 888 comparisons (25%) are unusable as such for lnRR. This afternoon we set them aside — and say so.

Trap 3 · InRR and g : same direction, different magnitudes



The 664 usable comparisons, computed both ways:

- Same **direction** always (both follow $\bar{X}_T - \bar{X}_C$)
- Similar **ranking** (Spearman $\rho = 0.84$)
- But very different **magnitudes**: g explodes when the SD is tiny

g depends on the SD (trap size, plot size, subsamples); $\ln RR$ does not. **Never mix both** in one analysis, and do not convert $\ln RR \leftrightarrow g$.

Other kinds of data, in one slide

Yes/no outcomes

Plants infested or not, seedlings alive or dead → a 2×2 table of counts → **log odds ratio** (or log risk ratio, the log of the ratio of proportions).

`escalc("OR", ai=, bi=, ci=, di=)`

Correlations

e.g. habitat cover vs pest control → **Fisher's z**. Pool **z**, not *r*, then back-transform.

`escalc("ZCOR", ri=, ni=)`

Conversions between metrics

Possible ($\log OR \rightarrow d, r \rightarrow d$), but only for a **minority** of studies: flag them and run a **sensitivity analysis** without them. **Never $\ln RR \leftrightarrow g$** . Formulas in the formula sheet (Borenstein et al. 2021 chap. 7).

Quick quiz

1. Earthworm biomass (g/m^2) under conservation vs conventional tillage, studies in many different units. Metric?

InRR — positive, proportional, unit-free.

2. $\text{InRR} = -0.36$. Change in %?

$100 \times (e^{-0.36} - 1) = \mathbf{-30\%}$.

3. Proportion of fields where a weed is present, with vs without cover crop. Metric?

log odds ratio (or log risk ratio).

4. Why use g rather than d ?

d **overestimates |effect|** in small samples; g corrects it.

KEY MESSAGES

- 1 | **Effect size + sampling variance** = the unit of every meta-analysis; the variance becomes the **weight**.
- 2 | The **data and the question** choose the metric: MD, g , lnRR, log OR, z . Fix it in the protocol.
- 3 | lnRR: check **Geary**, correct small-sample **bias**, handle **zeros** and **missing SDs** explicitly.
- 4 | Never mix **non-convertible** metrics (lnRR $\leftrightarrow$ g); flag and test any conversion.

Going further

All formulas (with the exact J , bias-corrected $\ln RR$ and its variance, conversions and their variances): see the **A4 formula sheet** handout.

- **Variability as an outcome**: $\ln CVR$, the log ratio of coefficients of variation (Nakagawa et al. 2015).
- **Paired / before–after designs** (SMCC, ROMC in *esca1c*) need the correlation between measurements.
- **Several treatments vs one control** create **dependent** effect sizes → multilevel models, this afternoon.

Extra, if time remains

Can we just average the percentages?

Same 440 pest comparisons, two “averages” (unweighted):

- mean of the % changes: **+33%**
- back-transformed mean lnRR: **-25%**

Why so different?

A % change is **asymmetric**: halving gives -50%, doubling gives +100%. A few large increases pull the mean of percentages up. Average on the **log scale**, back-transform at the end.

Which effect size?

PAIRS, 3 MINUTES

For each case, which effect size would you use?

1. Presence or absence of a pest in 20 intercropped and 20 monoculture fields.
2. Wheat yield in t/ha, same unit in every study.
3. Species richness, with several monocultures at **zero**.
4. Correlation between landscape complexity and pest abundance across farms.

1 · log **odds ratio** ("OR") · 2 · **mean difference** ("MD") or lnRR if you want a relative effect · 3 · **SMD** ("SMD"), lnRR is undefined · 4 · **Fisher's z** ("ZCOR")



In the online notebook

Each point of this session developed further, with the same data, more pitfalls and exercises with solutions:

- Effect sizes
- Formula sheet (appendix)
- From research question to estimand
- Dependence: map it before you model it

literaturesynthesis.github.io/notebook

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